## Module 15

# Training Modalities

## ❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Types of training categories
    - Active rest/recovery
    - Endurance
    - Threshold
    - Intervals
    - High-intensity intermittent training (HIIT)
        - 80/20 training practice
        - Integration of HIIT
    - Multiple workout training days
- Training environments
- Sport-specific drills
- Proper progression

## ❖ TRAINING CATEGORIES

Repeats, fartleks, HIIT, base … what do these things have in common? They are all specific workout/training types. While there are near-limitless names for workout types, training sessions fall under six primary categories:

1. **Active Rest/Recovery**
2. **Endurance**
3. **Threshold**
4. **Intervals**
5. **High-Intensity Intermittent Training (HIIT)**
6. **Multiple-Workout Training Days**

It is important to know that many athletes and coaches place an overemphasis on one or more of the aforementioned areas. Any emphasis on a particular training area (i.e., HIIT) should correlate with a specific point in the training process. For example, when focusing on building fitness and a mileage base, endurance training should be the primary focus, not HIIT.

In regard to runners who are considered well trained, it is advised to take a polarized approach to training, meaning all of the aforementioned training categories are combined in the training program/process. A 2014 study by Stöggl and Sperlich found that among well-trained endurance athletes (runners, cyclists, triathletes, cross-country skiers), a polarized training program that integrated high-volume training, threshold training, and high-intensity intermittent training (HIIT) improved endurance performance the most (697). This polarized approach increased endurance performance to a greater extent than training each training area independently.

As the subjects in this study were well trained and therefore already had a solid base aerobic fitness level, deconditioned individuals and those new to the sport of running would likely find a polarized approach too aggressive.

In summation, to be considered a complete and effective program, all of the preceding categories should be concurrently trained by well-trained runners, albeit at varying intensities and volumes based on the individual and the point in the program.

## ACTIVE REST/RECOVERY

While recovery can be a total rest day, active rest is often prescribed as a way to keep the muscles “loose.” These workouts are characterized by extremely low intensity for relatively short amounts of time.

**Terminology:** run/walk, jog, shakeout run (if before an event), recovery run

**Run/Walk**
Run/walk workouts are great for beginner runners as well as seasoned runners coming back from injury or doing a recovery workout. This type of workout combines running and walking and is up to you to determine the total time and ratio between running and walking.

**Shakeout Run**
These runs are done either the day before or the day of (a few hours before) a race. They are characterized by an easy pace (i.e., jog) and relatively short in distance. They are done for the purpose of loosening up the legs. If done directly prior to a race, it is considered some or all of the warm-up.

**Cross-Training**
Activities such as cycling and swimming can also be good for recovery, so long as the intensity is relatively low and the individual is used to the mechanics of the activity.

# ENDURANCE TRAINING

The purpose of endurance workouts is to increase muscular endurance and aerobic capacity. They are characterized by exercising at a low intensity (sub LT) and for relatively long periods of time. What constitutes a 'long time' is subjective and based on an individual. The pace during endurance sessions is typically steady with few fluctuations in intensity or speed.

**Terminology:** long steady distance (LSD), base, long run

## Long Run
LSD running is generally structured around a slower pace and longer distance. The distance is based on where a client is in the training program.

A high volume of training is often associated with an increased chance for injury. This is most correlated with total weekly training volume and dramatically increasing the volume during a training session. A 2012 study by Quinn et al. found that while muscular damage did occur during long runs, so long as an individual was adapted to this distance in regard to a progressive training program, there were no negative effects on a runner's running economy at submaximal speeds (688). Therefore, a runner who is well adapted to a training program should be able to adhere to the program in the days following a long run without negative consequences.

### Base Run
These are used to help a client adapt to running from a musculoskeletal standpoint as well as gain cardiovascular fitness. These runs are typically done at a comfortable pace. The distances of base runs increase as a client adapts to running and gains cardiovascular fitness.

## Intensity vs. Volume

Many runners try to get away with training as little as possible for their goal race. There are likely many factors for this, such as other time commitments, fear of injury due to overuse, boredom, etc.

While some of the aforementioned factors may be valid, the compensatory action of reducing training volume is typically to increase intensity. This is a slippery slope. Remember, the aerobic system adapts at a faster rate than the musculoskeletal system. Therefore, although an individual may be able to "handle" an intense cardiovascular workload, it is likely that the person's musculoskeletal system cannot.

While there is some physiological carryover between aerobic and anaerobic-type exercises, they largely do not focus on the same thing. As such, swapping out endurance for intensity is not a shortcut that is advised. While there is nothing wrong with decreasing volume to a point, it must be done intelligently and you should have historical data on the client that would suggest that the proposed decrease in training volume would be sufficient for the event being trained for. However, even in this case, a drastic decrease in volume or a swap for high-intensity workloads is not recommended.

Laursen et al. note that while there is overlap between adaptations that occur at low and high intensities, the metabolic pathways to elicit these adaptations are likely different. Additionally, and more important, he notes that low-intensity training (base training) is largely a prerequisite to high-intensity training having an impactful and lasting effect on performance (738). In other words, low-intensity/high-volume training should precede low-volume/high-intensity training for the best results.

### Short-Term Studies
As will be discussed in the HIIT section, quite a few studies tout the benefits of high-intensity training over high-volume/low-intensity training. It is postulated by

Magness that because of rapid adaptations from high-intensity training versus the extensive time required to see low-heart-rate-training adaptations, high-intensity training has been recommended over low-intensity training as a more efficient way to train for endurance events (520).

### Conclusions

While high-intensity/low-volume training may initially elicit faster physiological adaptations than low-intensity/high-volume training, it is advised to perform low-intensity training at medium to high volumes prior to implementing high-intensity workloads.

## THRESHOLD TRAINING

The purpose of threshold training is to increase one's lactate threshold (LT). This training occurs in a range slightly below to slightly above one's LT. While threshold training can be steady efforts across a relatively long period, it can also be shorter or have variations of intensity.

**Terminology:** Negative splits, tempo, time trial, tempo, progression runs

### Tempo

Tempo workouts consist of long time durations at or near race pace. Tempo efforts are typically a minimum of five minutes and are steady-state efforts at or slightly above/below LT (317). If above LT, the pace is around one's MLSS.

These workouts are typically performed just below race pace and are done for five to 20 minutes (excluding warm-up/cooldown), depending on the length of the event being trained for. A warm-up is recommended when performing tempo efforts.

### Time Trial

This is essentially a tempo effort but is typically done over a set distance and often with a set time goal in mind. Time-trial efforts are typically longer than a *tempo* workout. It is common for a time-trial effort to be anywhere from five to 10 miles long. They are done at or near MLSS.

### Progression Run

This is a run that starts at a particular pace and ends at a faster pace. For example, a six-mile run might start with the first four miles at an eight-minute-mile pace, the fifth mile at 7:30/mile pace, and the last mile at 6:45/mile pace. The exact structure is up to you. However, the constant is an escalating pace throughout the workout. This type of workout or race strategy is also referred to as *negative split*. This refers to running the second half of a run at a faster pace than the first.

## INTERVALS

Intervals are characterized by predetermined durations of intense cardiovascular efforts that are at or above threshold level and timed recovery periods. Recovery periods typically do not allow for full recovery.

The purpose of intervals is to increase aerobic capacity, increase LT, and improve recovery time. Interval intensity can be measured by things such as pace, time, distance, and heart rate. While an interval effort can be short or long, it is advisable not to have an effort be shorter than 30 seconds and not longer than five minutes. Efforts longer than five minutes would likely need to be done at a reduced intensity that would diminish the physiological benefits of the interval set.

Intervals have two primary components: intensity and time. Some intervals might increase in time as well as intensity, whereas other interval sets might increase in time but decrease in intensity. The structure of an interval set is dependent on what you deem most important for the progression of your client.

> Intervals done at a very high intensity and for short durations could be considered a high-intensity interval (HIIT) workout due to the high intensity and short rest periods.

**Terminology:** repeats, speedwork, pyramids, fartleks, surges/pickups

While the structures of intervals vary, five primary types are most applicable to running.

## Straight

Efforts and recovery aspects remain the same throughout the session.

- Ex: Three minutes at 150 beats per minute (bpm) followed by one minute of rest. Repeat this three times then rest for five minutes. Repeat cycle two times.

  **TERMINOLOGY**

  **Repeats:** These are characterized by running at a relatively fast pace for a set amount of time (or distance), recovery, and then running the same distance and/or pace again (e.g., run mile at seven-min/mile pace, easy jog quarter mile, run mile at seven-min/mile pace; repeat cycle two times).

  **Float:** Float runs are essentially *repeats*, but instead of an easy jog in between efforts, the pace is still relatively high. Float workouts train the body to be able to respond to midrace surges while not dipping below race pace. An example of a float run would be a track workout with alternating 400-meter efforts. The alternating efforts would be at race pace and 30 seconds over race pace. Typically a float workout is assigned a set number of efforts. If an individual can no longer maintain the assigned pace, the workout ends.

## Ascending

Efforts increase in duration throughout the session.

- **Example:** One mile at 8:00 pace, one-minute recovery jog > two miles at 8:15 pace, two-minute recovery jog > three miles at 8:30 pace, five-minute recovery jog; then repeat cycle.

## Descending

Efforts decrease in duration throughout the session.

- **Example:** Three minutes at 7:00 pace, two-minute jog > two minutes at 6:00 pace, one-minute jog > one minute at 5:30 pace, five-minute jog; then repeat cycle.

## Pyramid

Effort durations increase then decrease to the starting level.

- **Example:** 400 meters at 75 seconds, 200-meter recovery jog > 800 meters at 2:30, 400-meter recovery jog > 1600 meters at 5:00, 400-meter recovery jog > 800 meters at 2:30, 400-meter recovery jog > 400 meters at 75 seconds.

### Unstructured

This type of interval is characterized by hard efforts and recovery periods, but the durations, intensities, and recovery times vary.

#### RUNNING TERMINOLOGY

**Fartlek:** This is a Swedish word that means *speed play*. Fartleks are unstructured intervals. This means that a client randomly picks up the pace at random times throughout a workout. The duration of each "pickup" is also random. The only structure normally associated with fartleks is that your client might predetermine how many efforts to do during a session. An example of a fartlek would be upping the pace from one telephone pole to the next, but this is typically determined while running, not in advance.

**HILL REPEATS**

A common interval-training workout is hill repeats. As the name suggests, this workout consists of an individual running up a hill at a particular pace, resting (usually by running back down the hill), and then running back up the hill.

Hill repeats are often performed to become more efficient at running uphill, as well as to work on one's cardiovascular conditioning.

## HIGH-INTENSITY INTERMITTENT TRAINING (HIIT)

There has been a recent trend of training at relatively low volume and high intensity for endurance events. The diminished training volume is typically replaced by increased intensity (often termed *high-intensity intermittent training* or HIIT) (427).

HIIT is characterized by an intensity level that is equal to or greater than that of intervals. While the time frame for bouts of intensity/recovery might be close to that of intervals, HIIT often incorporates varying modalities of training (e.g., rowing, strength training, plyometrics, etc.)

HIIT has received plenty of accolades as well as criticism. Reasons why high-volume training might not be the best way to train for endurance events are:

- High-volume training wastes time as too much emphasis is placed on base-level training
- Rate of injury increases because of the high volume (overuse injuries)
- Boredom
- Many physiological adaptations that occur with traditional endurance training can be gained via high-intensity/short-duration workouts (328)

Critics of HIIT counter with the following arguments:

- If training for a long-distance event, there is no substitute for long-endurance training days/time periods
- Even if physiologically HIIT training was effective for training for long-endurance events, it does not address the psychological stresses of long-duration endurance efforts
- A solid base-training phase is required for the proper physiological adaptations to occur
- HIIT training is not applicable for all populations

*CrossFit® Endurance (CFE)* is one type of HIIT that has received a lot of attention. Founded by Brian MacKenzie, CFE is an offshoot of the popular workout *CrossFit®* (329). CFE emphasizes the key principles of HIIT and adds strength training as a major component, primarily through the practice of Olympic lift-type exercises and movements. CFE does not utilize a periodization-based training model (329).

The most cited and documented study done on the purported benefits of HIIT was performed by Tabata et al. The 1996 study has gained attention because of its findings related to the benefits of short, intense exercise on all areas of the body (170). Many people use the term *Tabata training* interchangeably with *HIIT*.

## Tabata Study

Tabata's 1996 study used a cycle ergometer and took place over a six-week period (170). Subjects were divided into two groups: moderate and high intensity. The study sought to assess improvements in anaerobic capacity (based on maximum accumulated $O_2$ deficit) and $VO_2$ max.

- The first group performed a steady-state effort at 70 percent of their $VO_2$ max for 60 minutes five days/week.
- The second group performed seven or eight sets of 20 seconds of anaerobic activity with 10 seconds of rest between each 20-second interval. This was also done five days/week for the six-week duration.

**Results:** The first group realized an average increase in $VO_2$ max of five ml/kg/min with no discernible increase in anaerobic capacity. The second group (HIIT) realized a $VO_2$ increase of seven m/kg/min and a 28 percent increase in anaerobic capacity. The conclusion of the study was that through high-intensity training, an individual might be able to see substantial increases in both aerobic and anaerobic capacity, likely because of the intense stimulus of both energy systems.

While there are various protocols for HIIT, the common denominator is extremely high intensities with short recovery periods.

Research on HIIT suggests that it is superior to traditional endurance training (low to mid intensity over long periods) in that subjects realize endurance and aerobic gains with a substantial decrease in time requirement. A 2006 study by Gibala et al. (174) divided a group of 16 men into two equal groups. One group performed 10.5 total hours of traditional submaximal endurance training over a two-week period, while the other group performed 2.5 total hours of HIIT. Both groups divided their training into six sessions. The subjects were tested pre- and post-training. The post-training findings indicated that both groups showed similar physiological adaptation levels in regard to increases in $VO_2$ max, glycogen levels, and hydrogen ion muscle-buffering capacity. The HIIT group also increased *anaerobic* capacity. These findings represent a huge opportunity for runners with limited time to train.

Despite the studies noted here, HIIT is not as well researched as many other areas of conventional endurance training. Therefore, it can be assumed that there are still many unknowns regarding the effectiveness of HIIT on distance running training, especially in respect to training for half and full marathons.

It is strongly advised to begin a training program with a solid base training phase that focuses on low heart rate training. This does not mean HIIT cannot be implemented during this phase. However, it should be done in moderation.

Depending on the training modality, HIIT can place increased stress on the muscular system. Stress on muscles and connective tissue must be gradually progressed to ensure that they can handle the increased workload.

Assuming that HIIT could fulfill all cardiovascular, musculoskeletal, and endurance needs of a runner, the one variable that it cannot take into account is the psychological aspect. The time it takes to complete a marathon typically ranges from three to six hours, with the sub-three-hour range typically reserved for professional and top age-group competitors. Therefore a distance-running program made up almost exclusively of HIIT-type programming has a high probability to leave a runner ill-prepared for the psychological demands of long-distance running, such as a marathon. This aspect cannot be ignored or underestimated.

## Effect of HIIT on Well-Trained Endurance Athletes

One study of particular interest examined the effect of HIIT on well-trained endurance athletes versus sedentary individuals and recreational athletes. A 2002 study by Laursen and Jenkins found that submaximal training on well-trained athletes did not seem to increase their endurance capacity or $VO_2$ max (171). It is theorized by Laursen and Jenkins that the only way to elicit additional endurance and physiological gains in well-trained endurance athletes is through HIIT. Interestingly, studies examined by Laursen and Jenkins found that HIIT done by well-trained athletes did not increase their oxidative or glycolytic enzyme activity. Therefore, they theorized that an increase in the muscle's ability to buffer hydrogen ions (delay muscle burn) was a possible reason for an increase in endurance capacity in well-trained endurance athletes.

Several studies have demonstrated the physiological benefits of HIIT on endurance athletes who had not previously integrated HIIT into their training program (225, 226, 227). For these athletes, the benefits of HIIT were realized quickly. Conversely, increased intensity of HIIT sessions in athletes who were already performing HIIT did not show significant increases in cardiovascular adaptations (228, 229).

## 80/20 Training Practice

A 2010 review by Stephen Seiler regarding subthreshold and high-intensity programs and their effect on participants found that programs comprising 80 percent light intensity (endurance-based program) and 20 percent threshold/high intensity were common and elicited positive results (238).

Seiler's review of research found that by incorporating two interval sessions per week, performance increased 2–4 percent in well-trained athletes (228). Seiler's research found that additional bouts of HIIT sessions past two times a week did not improve performance and, in some cases, led to the subjects having symptoms of overtraining (239, 240). Seiler believes a conflicting relationship between low- and high-intensity workloads is likely exaggerated. This is substantiated with the documented effectiveness of 80/20 training programs.

## Effect of HIIT on Metabolism

Studies by Talanian et al. (2007) and Perry et al. (2008) both had similar findings in regard to HIIT's effect on fat metabolism. The Talanian study found that HIIT sessions over a two-week period resulted in the subjects (eight moderately active women) realizing a 13 percent increase in $VO_2$ max and a 36 percent increase in whole-body-fat oxidation (172). The Perry study used untrained, active subjects and asked them to perform HIIT sessions for six weeks (three times a week at one hour per session). Subjects in this study increased their fat oxidation by 60 percent (173).

## Conclusions

There is no doubt that HIIT is a time-effective way to get a great workout and should be integrated into a running program. The standpoint of UESCA is that clients still need to undergo a base training phase, and that the length of the phase should be based on the physiological adaptation of the client, not necessarily a predetermined time frame.

- It is up to you to experiment with HIIT integration to see how each client responds. What might be the correct training volume for one person might not be the right volume for another.

- Based on research regarding HIIT, it appears to offer individuals a way to gain many of the same benefits of traditional endurance training in a fraction of the time. In regards to the Gibala et al. study (174), the training time is approximately one-quarter that of traditional training to elicit the same physiological responses.

- Integration of HIIT principles into a training program is recommended, unless a client does not have a solid base fitness level, in which case HIIT should be done sparingly. While HIIT is highly effective, it does not eliminate the need for endurance-based sessions that are focused on low to medium intensity.

- HIIT principles are great for clients who have limited time to train.

- It is likely that an 80/20 distribution of low and high intensity, respectively, will elicit positive results.

- The exact intensity and duration of HIIT intervals and rest periods are customizable.

- More statistically significant research is needed to assess the accuracy and validity of a HIIT-based program for endurance-sports-based individuals.

- During a HIIT session, the oxidatitive energy system is trained during the rest periods because of *excess post-exercise oxygen consumption* (EPOC). As noted in *Module 4*, the oxidative energy system works very hard to bring an individual back to baseline after intense anaerobic work. In other words, the oxidative system works hard to repay the oxygen debt to return the body to equilibrium (homeostasis).

# MULTIPLE-WORKOUT TRAINING DAYS

## Two-a-Days (TAD)

A TAD incorporates two workouts within a single day. There are some basic guidelines of TADs.

- No back-to-back TADs
- Must have at least eight hours between workouts
- TADs should be performed only by runners who have a solid mileage/fitness base

### Common TAD Structures

- One of the TAD workouts is a running workout while the other workout is some form of cross-training (e.g., swimming, weight training, cycling).
- One of the TAD workouts is cardiovascular focused while the other workout is focused on strength, recovery, or skill/form.
- Because of the mileage requirement, TADs are often performed by runners who are training for ultra-marathons. In these cases, typically both workout sessions are mid-long-distance runs at submaximal efforts.

Two recent studies suggest that training twice a day compared with training the same total volume (sum of the two sessions/day) every other day induces greater physiological adaptations (242, 243). Therefore, TADS should not be viewed solely from a perspective of convenience, but also from a perspective of performance enhancement.

# PROGRAMMING INTENSITY

Introducing and/or increasing intensity in a training program requires a slow and disciplined progression. Training modalities such as sprints, hill repeats (for speed), and intervals require a runner to produce more power than under normal running conditions. This increase in power typically occurs in conjunction with an increase in leg turnover, and therefore the running stride becomes more ballistic. Additionally, these training modalities often change the mechanics of a runner – specifically increased hip extension and transverse hip rotation.

## BIOMECHANICAL ADAPTATIONS

Given the likely increase in hip extension and transverse hip rotation, if one's body is not adapted to an increased range of motion, injury is a likely result. While a "tight" muscle(s) may assist a runner in regard to the stretch-shortening cycle (SSC), the muscle(s) must have full range of motion. In addition to gradually increasing the speed and quantity of the aforementioned training modalities, performing stretches that target the hip flexors, hip adductors (groin muscles), and abdominal muscles is advantageous.

You must inform your clients to take stock of their body during and after speedwork sessions. If they feel pain, excessive muscle tightness, or limited range of motion, they must stop to minimize the chance for injury.

## BUILD UP SPEED

Introducing speedwork and intensity into a program must be done in small amounts and with small speed increases. The theory *walk before you run* is applicable to speedwork as there is a natural hierarchy to the type of workouts that are introduced in a training program. For example, prior to introducing speedwork a client should already be performing threshold training (tempo runs, progression runs) and fartleks. While not pure speedwork, these training modalities help the body adapt to running at a faster pace.

The first interval-training session might look something like this:

- Two-mile warm-up at eight-min/mile pace
- 3 x 200 meter efforts at seven-min/mile pace with a 400-meter jog in between
- One-mile cooldown jog

As the runner progresses, the interval-training session might look like:

- Two-mile warm-up at eight-min/mile pace
- 4 x 400 meter efforts at six-min/mile pace with 200-meter jog in between
- One-mile easy jog
- 4 x 200 meter efforts at 5:30-min/mile pace with 200-meter jog in between
- One-mile cooldown

## CONTROLLED ENVIRONMENT

Performing ballistic exercises that simulate the range of motion during speedwork (especially hip flexion, extension, and transverse rotation) is a great way to adapt the body to the stresses and range of motion of speedwork but in a controlled environment. Exercises such as jump lunges, vertical/lateral jumps, and torso rotations with a medicine ball are great to help adapt a runner to the stresses and range of motion of speedwork. These exercises need to be progressed properly as well.

# ❖ TRAINING ENVIRONMENTS

Course specificity is an overlooked aspect of program design. Race course terrain and characteristics need to be incorporated as much as possible in training. Ideally, the website of a race will post the race profile detailing the elevation. This will help determine what the training environment for your client should consist of. If the course profile is not online, it would be prudent to email the race director to get this information. By knowing the course profile, you can tailor specific workouts and race strategies.

Following are five primary types of training environments, each with its own advantages and disadvantages:

1. **Track:** A track is a great training environment for speedwork and, more specifically, interval-type training sessions. The inside of the inner lane (lane 1) is 400 meters long. Therefore, the distances of outer lanes are greater than 400 meters. Most tracks are made of cinder or a rubber-type compound.

    - **Disadvantages:** Boring, can be difficult to get track time if already utilized by a team or school.

2. **Road:** Often has varied terrain.

    - **Disadvantage:** Often puts the runner far from home.

3. **Trail:** Running off-road offers many benefits over running on road. Advantages are greater shock absorption, varied terrain, and an uneven

surface that challenges a runner's balance and ankle/leg stability. Trail runs and cross-country races are run almost exclusively off-road. Additionally, many ultra-marathons integrate off-road sections for some or the entire course.

- **Disadvantages:** Unstable surface; if in a hunting area, do not run during hunting season; unpredictable conditions.

**Trail Runs/Cross-Country**

If your client is going to participate in a race in which some or the entire course is off-road, the individual will need to do specific off-road training.

Running off-road requires much greater ankle stability and proprioception/kinesthetic awareness than running on the road. Without specific training, an individual greatly increases the chance for injury. As with implementing any new training element, a slow and progressive adaptation to trail running must occur.

In addition to incorporating trail runs into a training program, performing ankle-stability exercises and ballistic movements in the frontal plane can help an individual adapt to running off-road.

4. **Treadmill:** A treadmill offers many advantages over running outdoors. Primary advantages are increased shock absorption, not weather dependent, controlled environment (control over speed and incline), and often a more time efficient workout.

   - **Disadvantages:** Boring, treadmill belt artificially enhances gait, thereby changing the leg muscular activity.

When running outside, the client must be aware of the surroundings at all times and know the route and potential hazards.

## Treadmill vs. Running Outside

In many respects, running on a treadmill is the polar opposite from running outside. When running outside, an individual must push off the ground with the feet, whereas when running on a treadmill, a runner is primarily moving the legs to keep up with the treadmill belt. Running on a treadmill requires a runner to push off the “ground” (i.e., belt) a little, but nowhere near the extent that is required when running outside. Therefore, the muscles and the way they are utilized differ between running on a treadmill and outside. While it is not advisable to run on a treadmill exclusively, it does have benefits. Running inside is the best way for a coach to assess a runner’s gait. As treadmills tend to have better shock-absorption properties than outdoor running surfaces, for clients who are injury prone because of the impact nature of running, treadmill workouts can be quite beneficial.

A 2014 study by Kaplan et al. found that when running outside, the foot strikes the ground with 201 percent of one’s body weight whereas running on a treadmill results in 175 percent of one’s body weight (698). As a side note, exercising on an elliptical machine resulted in only 73 percent of one’s body weight. Therefore cross-training using closed-chain cardiovascular exercises such as ellipticals and bikes will likely reduce the chance for injury and increase recovery time over running. Nonimpact open-chain exercises, such as swimming and pool jogging, likely decrease the chance for injury as well.

### Why Does Treadmill Running Suck?

While not everyone thinks treadmill running sucks, a large percentage of runners find running on one harder than running outside.

As a treadmill has good shock-absorption properties and helps to propel a runner’s legs rearward, in theory it should feel easier than running outside. Additionally, the mechanics of treadmill running and running outside are the same (699) – so what gives?

A 2012 study by Kong et al. examined this exact phenomenon (700). In this experiment, runners ran on an outdoor track for three minutes and then immediately ran on a treadmill for three minutes, followed by another three-minute run outdoors. When running on a treadmill, subjects were instructed to run at the same perceived intensity and speed as on the track. The results showed that both of the outdoor runs occurred at the same pace while the treadmill run was performed at a substantially slower pace.

The study found that the likely culprit of this mismatch of perception of pace and speed is due to a distortion of visual inputs. In other words, the perception of increased difficult is purely psychological.

This finding correlates with another study that found the perception of distance relates significantly to the exercise outcome (701). The closer subjects are to a perceived target or finish, the faster they go and the easier the task is perceived to be. In this study, participants walked a course with no defined finish line and then walked the same distance with a set finish line. Participants noted feeling more fatigued when no finish line was present.

Because there is no physical finish line when running on a treadmill, it is likely more mentally exhausting and thus is perceived to be more difficult than running outdoors.

This information is important to take into consideration when working with clients who integrate treadmill running into their program and/or prescribing treadmill workouts.

An ancillary benefit to treadmill running is the auditory feedback the treadmill provides. Unlike the ground, a treadmill running deck is suspended and allows users to hear the impact of their foot strike. The louder the foot strike is, the greater the impact to the feet and legs. This auditory feedback allows a runner to modify the gait to become smoother and more efficient.

# ❖ RUNNING DRILLS

This section consists of running-form drills to integrate into a training program. This is by no means a complete list, but rather a comprehensive overview of popular drills and workout types. You will inevitably create new drills that will best serve your client.

> Some of the following drills are also noted in the biomechanical sport-specific modules.

## Run Uphill

The form used to run up a slight hill (5–8 percent grade) matches that used when learning how to run with a midfoot strike.

- The body leans slightly forward.
- The lead leg does not lift up excessively, but simply moves forward to support the upper body.
- The lead foot hits the ground with the midfoot, not the toes or heel.
- The lead foot lands when the body is directly over the foot, not in front of the body.

## Run Downhill

The tendency when running downhill is to elongate the running stride and to lean back into the hill. This is primarily done for the purpose of controlling one's speed (i.e., braking). The stride length should stay about the same throughout a run, regardless if the grade is flat, uphill, or downhill. Therefore the form when running downhill should not vary much from the form at any other time.

## Run with Slight Forward Lean

From the ankles to the top of the head, there should be a slight forward lean when running.

This is primarily for the purpose of the body being "over top" of the legs at the beginning of the drive phase. This allows for immediate forward propulsion. When the body is behind the legs at the beginning of the drive phase, the body must catch up to the drive foot in order for forward propulsion to occur. The other reason for a slight forward lean is to reduce the chance of the knees hyperextending via excessive lengthening of the hamstrings.

## Partner Pull-Back Run

Place a harness around your client's waist (can use a long theraband or padded rope) and stand behind the person holding the ends of the rope. Then have your client start running at a moderate pace. You will be running behind (braking), trying to reduce the client's speed. This restriction forces the client to have a slight forward lean and drive off the midfoot versus heel.

## Strides

Strides are essentially sprints (not 100 percent effort) that are done before and/or after runs/races. The purpose of performing them prior to a race is to warm up the legs to increase blood flow as well as to get the mind adapted to what race pace will feel like. Strides also work to enhance the neuromuscular efficiency of a runner.

When done at the end of long or easy runs, strides train the neuromuscular system to increase speed rapidly. This not only enhances one's form, but also one's speed – especially at the finish line.

Strides are typically done independent of a run. When increases in speed are integrated into a run, they normally fall into the category of *speedwork*, which strides are not. While strides increase one's heart rate, the primary physiological adaptation is neuromuscular, not cardiovascular.

If strides are done before an event, a runner must be sufficiently warmed up prior to performing them. To perform strides, a runner starts off running several steps at a jog then increases the pace quickly so that they are between 80 and 90 percent of their maximum speed within 15 to 25 foot strikes. Once this speed has been attained, the goal is to hold this pace for another 10 to 20 foot strikes before slowing down. A critical aspect of strides, including the deceleration, is to maintain as close to perfect form as possible. As strides train the body from a neuromuscular standpoint, performing strides with incorrect form trains the body to have incorrect mechanical patterns.

A client should do no more than five to 10 strides. If your client begins to feel tired from the strides, the individual has done too many.

There is not a lot of formal, peer-reviewed research pertaining to the efficacy of strides as it relates to increasing running performance. In fact, only one study has been found at this time.

As a result, the validity of strides is largely circumstantial as there are many variables that could increase a runner's performance, including those that pertain to increasing one's neuromuscular efficiency. The singular study is noted below.

## THE STUDY

A 2015 study by Barnes et al. found that runners who performed strides wearing weight vests (20 percent of BW) elicited the following results ten minutes after performing the strides (672):

- 2.9 percent increase in peak running speed
- 20.4 percent increase in leg stiffness
- 6 percent increase in running economy
- Slight decrease in cardiorespiratory demand

These results can all be attributed to an increase in leg stiffness, specifically in regard to the muscles of the lower leg and Achilles tendon.

Based on these results, the question as it pertains to conventional strides (not wearing weight vest) is whether they would elicit a meaningful percentage of the aforementioned results.

## Discussion

In lieu of a weight vest prior to the start of a race/run, unilateral plyometric leg jumps focused primarily on ankle plantar flexion would be an appropriate substitute.

## Conclusion

When warming up for a short- to middle-distance event (10K), a combination of unilateral plyometrics and standard strides are recommended approximately 10 minutes prior to the start of the event. Post-run or race, standard strides are recommended for the purposes of neuromuscular training. However, as noted above, more research in this area is needed.

# Fartleks, Pickups, Surges

It is likely that if you have been running for a while, you have heard of one or more of these terms. With the possible exception of small nuances between them, they are essentially the same thing – short increases in speed from an existing pace.

Three reasons for performing these workouts are:

1. Increase cardiovascular fitness
2. Adapt to changes in running tempo
3. Train to be able to respond to surges midrace

### What the Heck Is a Fartlek?

*Fartlek* is a Swedish term meaning *speed play*. Like pickups and surges, fartleks represent a fusion of steady-state running and intervals.

## ❖ SUMMARY

- The six primary training categories are:
  - Active Rest/Recovery
  - Endurance
  - Threshold
  - Intervals
  - High-Intensity Intermittent Training (HIIT)
  - Multiple-Workout Training Days
- The most cited and popular HIIT study is the Tabata study.
- Integration of HIIT principles is encouraged so long as a client has a solid base level of fitness.
- HIIT provides a time-crunched client a way to get in a great workout in a short duration of time.
- An 80/20 training practice refers to 80 percent of the training done at endurance-level intensity while 20 percent is done at high intensity (i.e., HIIT).
- When running on a treadmill, a runner is more or less keeping up with the moving belt. For this reason, it is important to not train solely on a treadmill.
- Runners often perform multiple workouts within the same day.
- Treadmill running is often perceived as being more difficult than running outdoors, likely because of the challenging mental aspect.
- Strides can likely assist a runner to become more efficient from a neuromuscular aspect.
- Sport-specific drills are an important part of a training process.